# ASSIGNMENT #03

**Course:** CHE 433: PROCESS SIMULATION WITH ASPEN PLUS
**Term:** Fall 2025
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**Student names:** ____________________________________

**Due Date:** Friday November 14th, my office (6:00PM)

## INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name(s) in the report.
- Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
- Upload your Aspen Plus Simulation file (*.apw)
- Bring a hard copy of your report as a PDF file by the Due Date and Time in class

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

To perform the task of creating a Bio-Oil using the initial 50,000 $^{kg}/_{hr}$ , we carry out a process involving pyrolysis.

As this process involves a number of units to design, they have been listed in the table below:

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| Name | Unit Operator | Purpose |
|---|---|---|
| DRY-100 | Dryer | To ensure the biomass is completely dry before entering the reactor |
| CYC-100<br>CYC-200<br>CYC-201A<br>CYC-201B | Splitter (SSplit) | Divides streams based on splits specified in the outlet streams |
| R-200 | Reactor (R-Yield) | This is the pyrolysis reactor it is a nonstoichiometric reactor based on a known yield |
| Mix-200 | Mixer | Mixes the streams coming out of the reactor, moisture, and fluid gas streams |
| E-200<br>E-202<br>E-203<br>E-204<br>E-206 | Heater | Increase the outlet temperature of the stream before going into the next unit operation |
| E-201<br>E-205 | Heat Exchangers | Transfer of heat from one medium to another to either cool or heat the streams |
| R-201A<br>R-201B | Reactor (Rstoich) | Based on known fractional conversions. With the first reactor being used for the actual reaction and the second reactor just for combustion. |
| ST-200 | Compressor | Increases the pressure of the streams. |
| SEP-200 | Separator (Sep2) | Separates components into two streams. Based on flows and purities. |
| Dec-200 | Decanter | Continuously separate 2 immiscible liquids |

Image of PFD

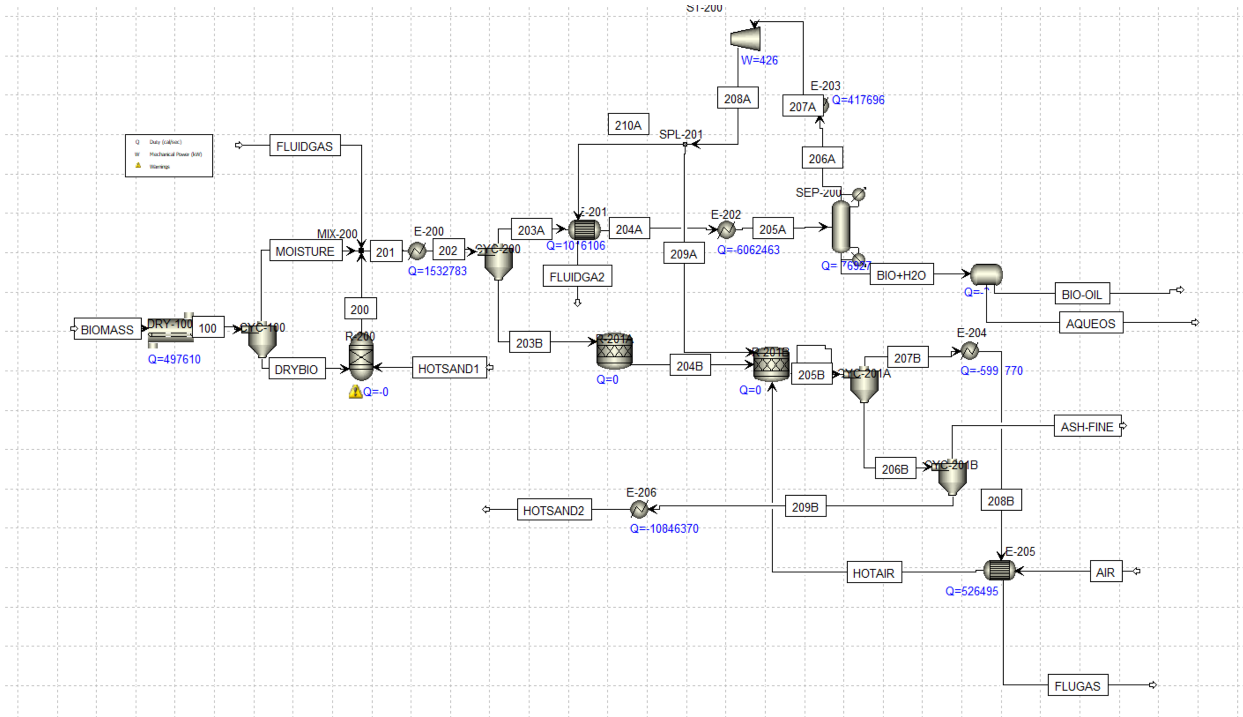


*Figure 1: PFD of process*

Explanation

The process begins with the Biomass stream being sent through a dryer unit with the specifications:

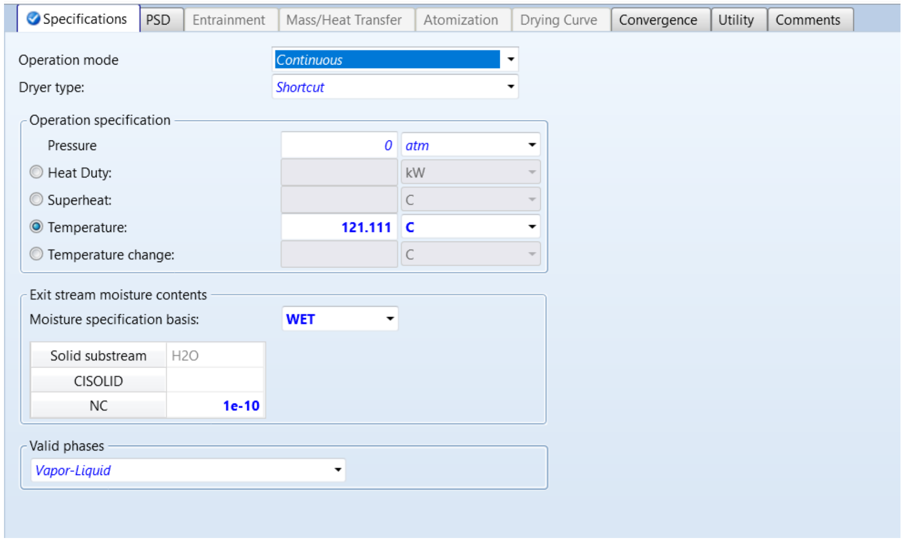


*Figure 2: Dryer Unit Specifications*

This enables the SSplit unit directly linked to the output stream of the Dryer unit to be able to discern between the moisture component and dry biomass coming from the feed, so that the only components going into the R-Yield reactor are the dry biomass.

The specification for the SSplit is shown below:

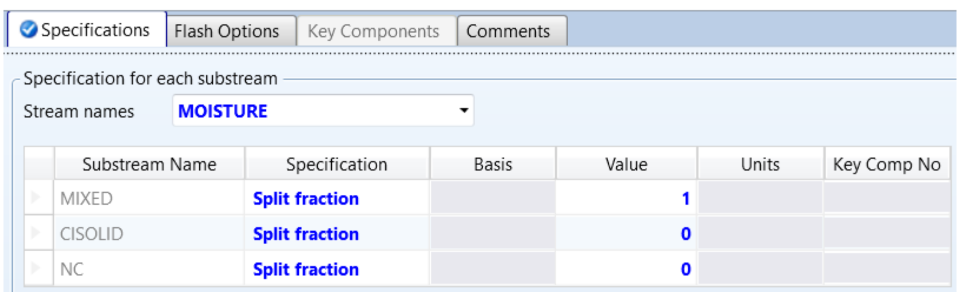


*Figure 3: Outlet MOISTURE stream specification of SSplit*

Thus, the only component in the MOISTURE stream is the one specified, $H_2O$. Now our R-Yield can process the dry components with ease, given that the yield specifications are in the problem statement. Along with this, there is an inert component: Sand, which enters the R-Yield reactor in order to deliver the energy required to carry out the pyrolysis reactions.

The R-Yield reactor specifications follow as listed:

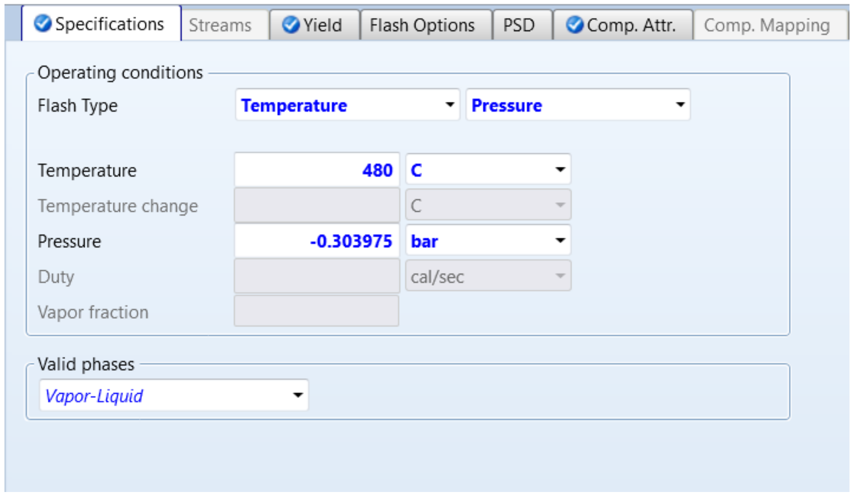


*Figure 4: R-Yield reactor Specifcations*

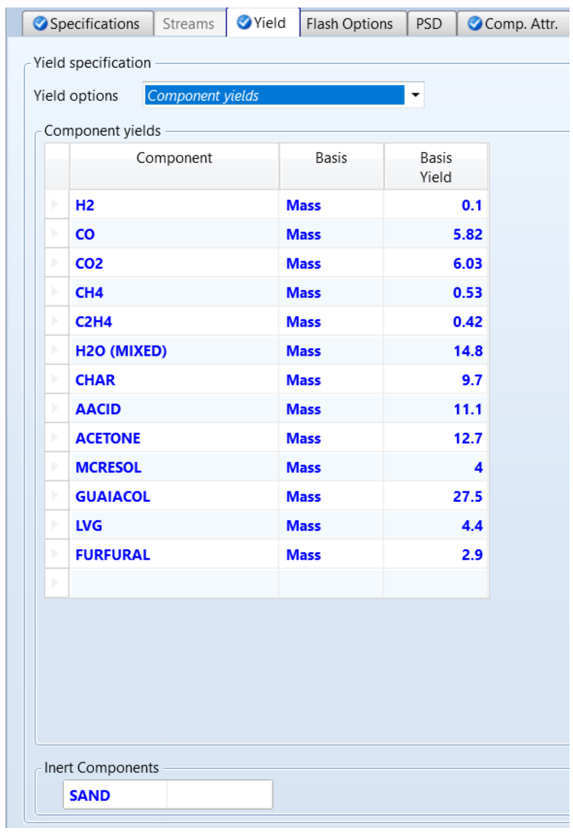


*Figure 5: R-Yield, Yield specifications*

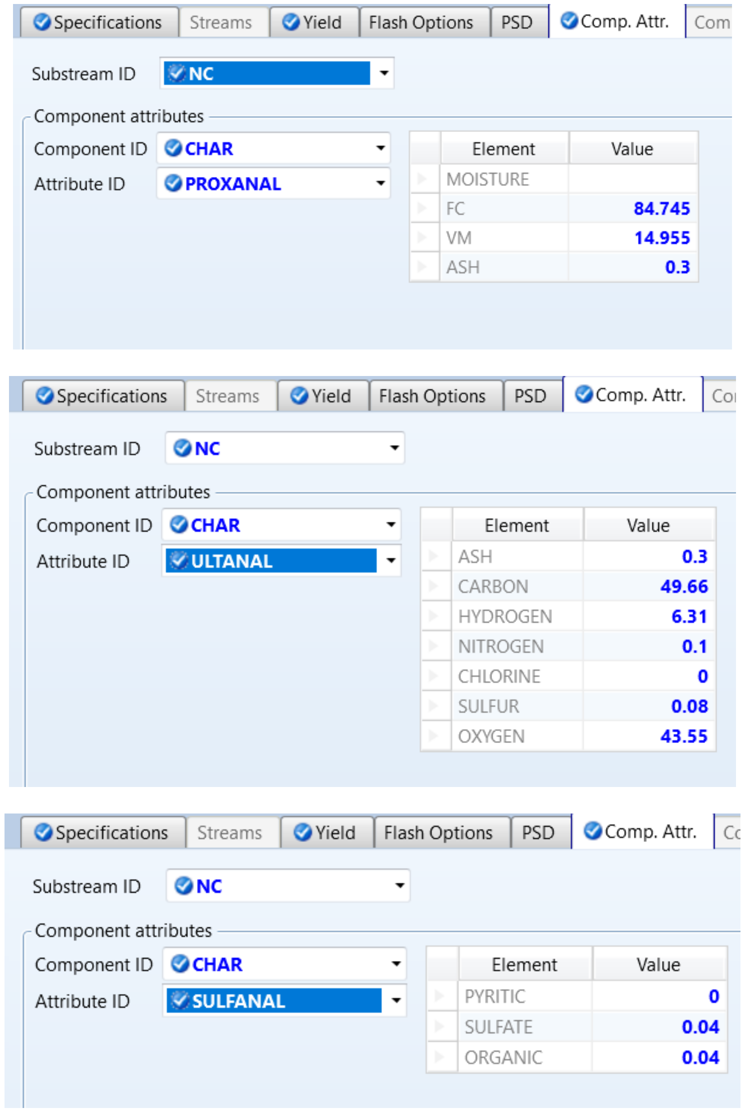


*Figure 6: Comp. Attributes for Char*

Once we get our products from the R-Yield, they are once mixed again with the once-removed Moisture component in the mixer (MIX-200). Here we also add a flue gas stream that essentially help to carry the solids in the streams and units.

The now mixed streams go towards a “fake” heater unit holding the specifications:

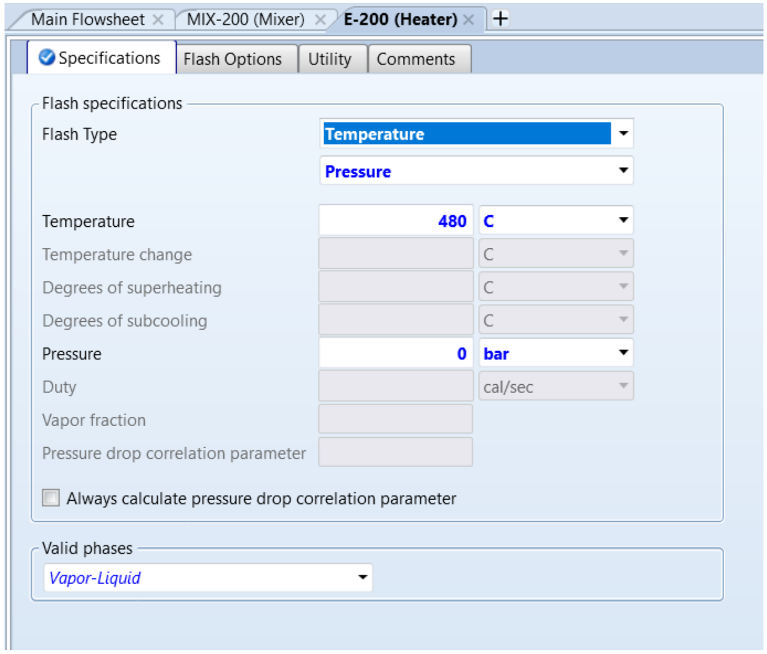


*Figure 7: Specifications of "fake" heater unit*

This stream is then sent into the SSplit unit (CYC-200) where the stream is once again split into biomass and the remaining components.

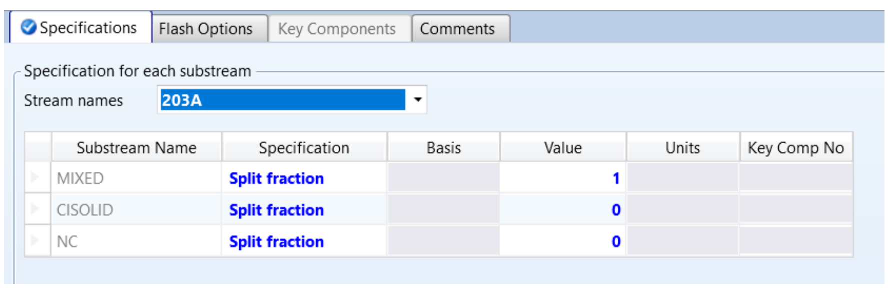


*Figure 8: CYC-200 specifications.*

The biomass (203B) is sent to the pyrolysis reactor where the breakdown of char into ash will be carried out. When specifying the reactor we initially put a coefficient guess of 1 for each component, however this is overwritten by the calculator (C-RXN) that takes the molar mass of the participating units and manages to find the number of moles required for the reaction. Stream 203B goes into our char combustor reactor.

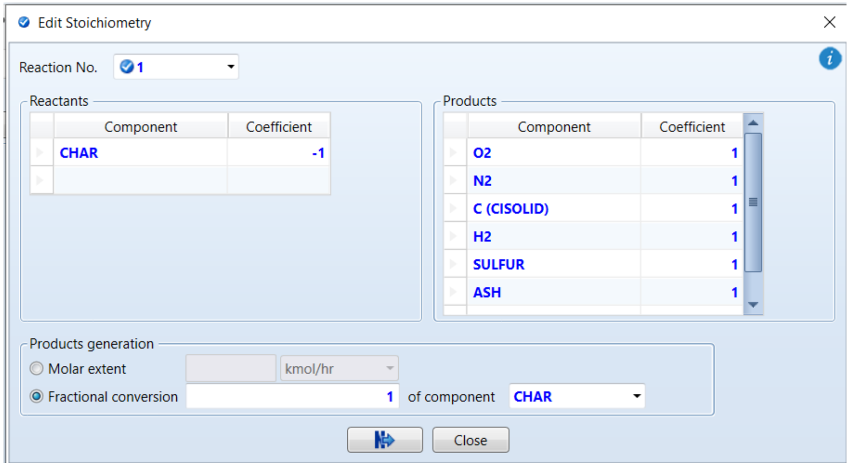


*Figure 9: Reaction taking place in R-201A*

| Variable | Information flow | Definition |
|---|---|---|
| MO2 | Import variable | Unary-Param Variable=MW ID1=O2 ID2=1 |
| MN2 | Import variable | Unary-Param Variable=MW ID1=N2 ID2=1 |
| MC | Import variable | Unary-Param Variable=MW ID1=C ID2=1 |
| MH2 | Import variable | Unary-Param Variable=MW ID1=H2 ID2=1 |
| MS | Import variable | Unary-Param Variable=MW ID1=SULFUR ID2=1 |
| NO2 | Export variable | Block-Var Block=R-201A Variable=COEF Sentence=STOIC ID1=1 ID2=O2 ID3=MIXED |
| NN2 | Export variable | Block-Var Block=R-201A Variable=COEF Sentence=STOIC ID1=1 ID2=N2 ID3=MIXED |
| NC | Export variable | Block-Var Block=R-201A Variable=COEF Sentence=STOIC ID1=1 ID2=C ID3=CISOLID |
| NASH | Export variable | Block-Var Block=R-201A Variable=COEF Sentence=STOIC ID1=1 ID2=ASH ID3=NC |
| NH2 | Export variable | Block-Var Block=R-201A Variable=COEF Sentence=STOIC ID1=1 ID2=H2 ID3=MIXED |
| NS | Export variable | Block-Var Block=R-201A Variable=COEF Sentence=STOIC ID1=1 ID2=SULFUR ID3=MIXED |
| MOIST | Import variable | Compattr-Var Stream=203B Substream=NC Component=CHAR Attribute=PROXANAL Ele... |
| ULTIMATE | Import variable | Compattr-Vec Stream=203B Substream=NC Component=CHAR Attribute=ULTANAL |

*Figure 10: List of Variables for calculation*

```
W0= MOIST/100
DRY=1-W0
NO2=DRY*ULTIMATE(7)/MO2/100.0
NN2=DRY*ULTIMATE(4)/MN2/100.0
NASH=DRY*UlTIMATE(1)/100.0
NC=DRY*ULTIMATE(2)/MC/100.0
NH2=DRY*ULTIMATE(3)/MH2/100.0
NS=DRY*ULTIMATE(6)/MS/100.0
```

*Figure 11: Fortran code for calculations*

This allows for the combustion reaction in the next R-Stoic reactor (R-201B) with the addition of a hot air inlet that is heated by E-205. Doing so allows us to separate the product stream into three components: Fluegas – which can be used to heat up the air inlet, Fine Ash – one of the products and finally, the hot sand we had used in the R-Yield reactor beforehand.

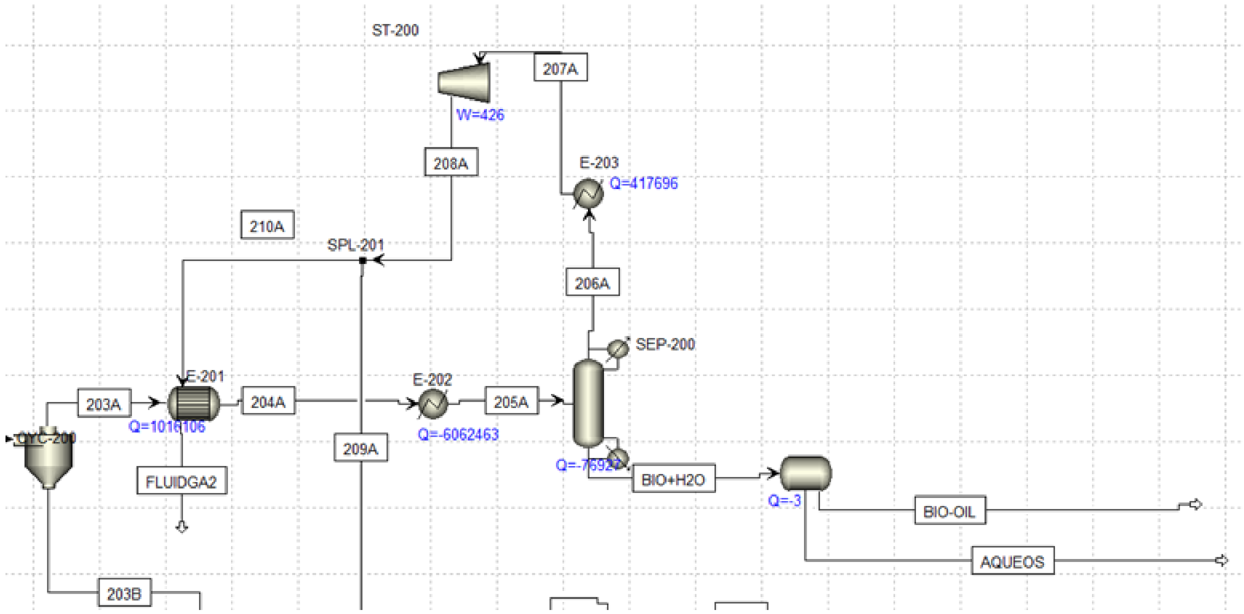


*Figure 8: Top section of splitter*

Looking at the top section of the splitter, we have the following PFD. After separating the mixed sub stream from the CISOLID and NC.

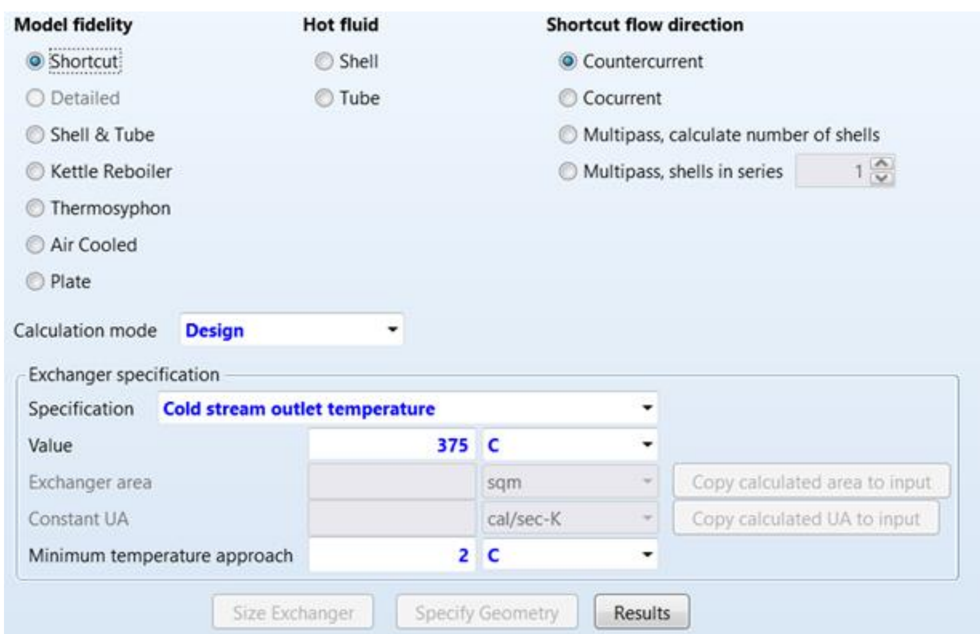


*Figure 9: E-201specifications*

E-201 is a heat exchanger that is capturing the heat from the hot stream of 203A in order to heat the fluidized gas before it enters the pyrolysis riser. This prevents cold gas from cooling the hot sand and dropping the pyrolysis temperature. Improving energy efficiency and keeping the reactor operating at the correct temperature. Now in our PDF we do not have FLUIDGA2 going straight into the pyrolysis reactor as when we attempted this ASPEN would crash every time. Which is why we have a separate FLUIDGAS stream going into MIX-100

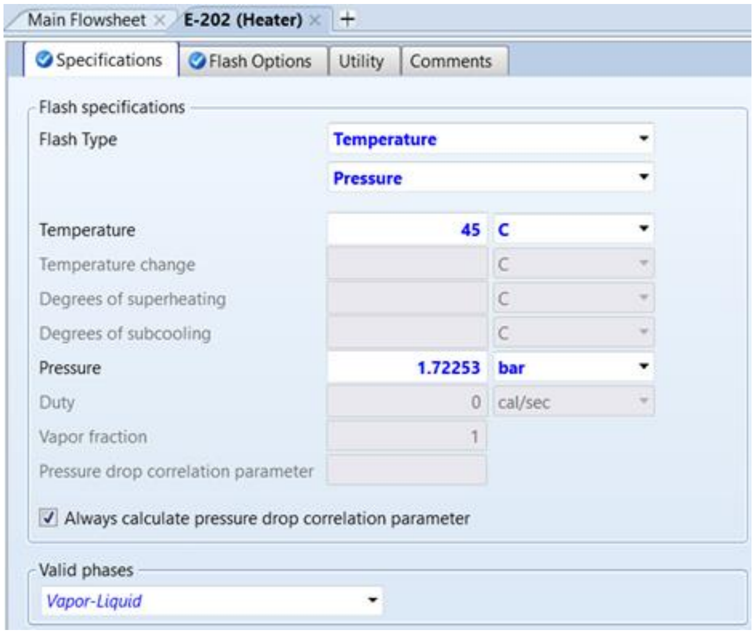


*Figure 10: E-202 Specifications*

Stream 204A is still hot coming out of the heat exchanger. With a Need to cool it down even further before entering the separator. Using a condenser with the design specs of 45°C and 1.7 bar as these are the conditions that the stream needs to be as given by the problem statement.

Main Flowsheet | SEP-200 (Sep2)

Specifications | Feed Flash | Outlet Flash | Utility | Comments

Substream: MIXED  Outlet stream: 206A

Stream spec: Split fraction

| Component ID | 1st Spec | | | |
|---|---|---|---|---|
| H2 | Split fraction | | 1 | |
| N2 | Split fraction | | 1 | |
| CO | Split fraction | | 1 | |
| O2 | Split fraction | | 1 | |
| CH4 | Split fraction | | 1 | |
| C2H4 | Split fraction | | 1 | |
| CO2 | Split fraction | | 1 | |
| SO2 | Split fraction | | 1 | |
| NO2 | Split fraction | | 1 | |
| ACETONE | Split fraction | | 0.96 | |
| 2MEFURAN | Split fraction | | 0 | |
| BENZENE | Split fraction | | 0 | |
| H2O | Split fraction | | 0.2 | |
| TOLUENE | Split fraction | | 0 | |
| AACID | Split fraction | | 0 | |
| CYPENTON | Split fraction | | 0 | |
| PXYLENE | Split fraction | | 0 | |
| FURFURAL | Split fraction | | 0 | |
| BZFURAN | Split fraction | | 0 | |
| INDANE | Split fraction | | 0 | |
| INDENE | Split fraction | | 0 | |
| 5MEFURAN | Split fraction | | 0 | |
| MCRESOL | Split fraction | | 0 | |
| GUAIACOL | Split fraction | | 0 | |
| MNAPTHEN | Split fraction | | 0 | |
| NAPHTHOL | Split fraction | | 0 | |
| CONALDE | Split fraction | | 0 | |
| LVG | Split fraction | | 0 | |

Specifications | Feed Flash | Outlet Flash | Utility | Comments

Inlet flash specifications

Pressure: 1.72253 bar

Valid phases: Vapor-Liquid

Flash options

Maximum iterations: 30

Error tolerance: 0.0001

*Figure 11: Specifications of SEP-200*

These are the specifications that we gave the SEP-200. In this separator we assigned split fractions to control exactly which components leave with each outlet stream. All permanent gases such as H2 , CO, CO2, CH4, N2. are given a split fraction of 1. This means that they fully exit with the vapor outlet because they can't condense. The heavy bio-oul compounds were all given a split fraction of 0. This makes sure they stay in the liquid outlet. Water is given an intermediate split aswell as acetone to show partial vaporization.

Coming out the bottom we have BIO+H2O which needs to get separated. Using a decanter with the following specifications.

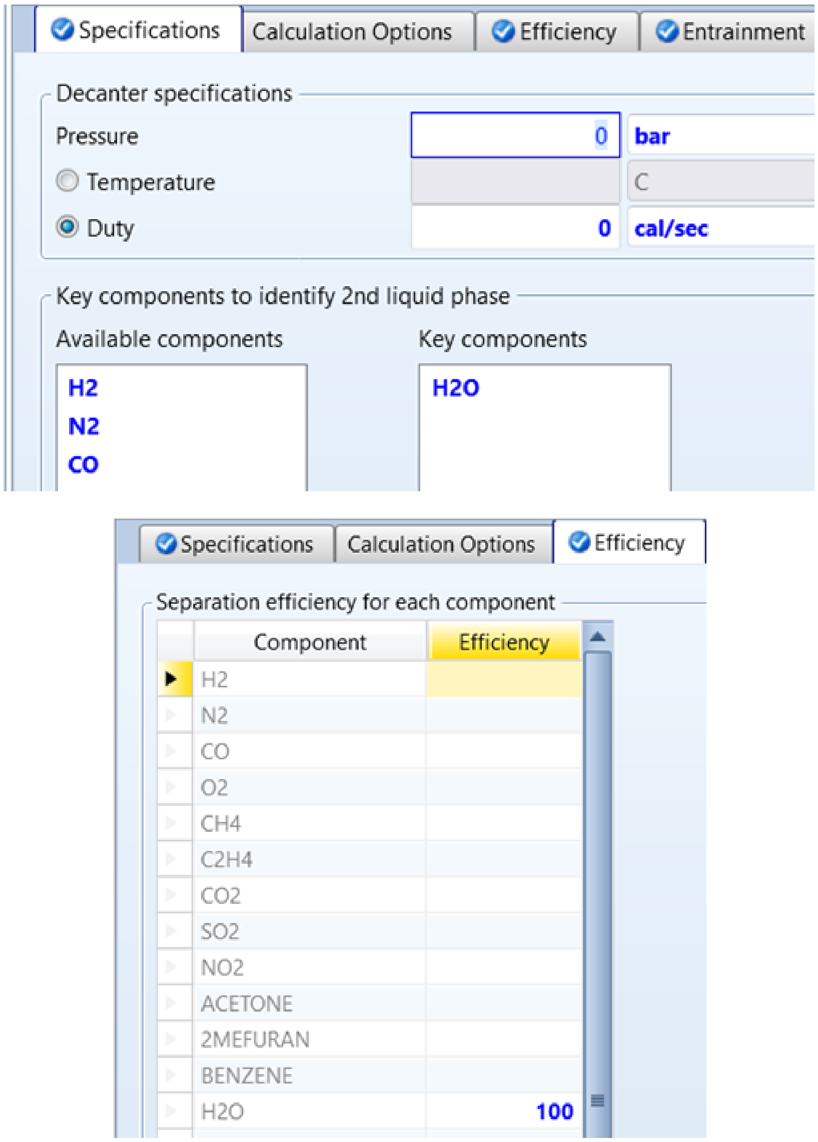


*Figure 12: Specifications for Decanter*

A decanter is used because the condensed liquid leaving separator 2 naturally splits into two immiscible liquid phases. Organic bio-oil and aqueous. Specifying H2O as the key component to identify the second liquid phase because water forms the aqueous layer from the organic bio-oil. The decanters separation efficiency for water is set to 100%. This makes sure that all the condensed water goes to the aqueous outlet and does not remain mixed with the bio-oil phase.

Coming out of the distillate there is E-203 and ST-200 which are a heater and compressor respectively. These are used to increase the pressure and temperature of the stream. Which proceeds to go into SPL-201. Stream 210A goes through the E-201 and gives us our FLUIDGA2. Stream 209A goes into R-201B, our second stoich reactor.

After looking over the entire process we have 2 more design specs set up

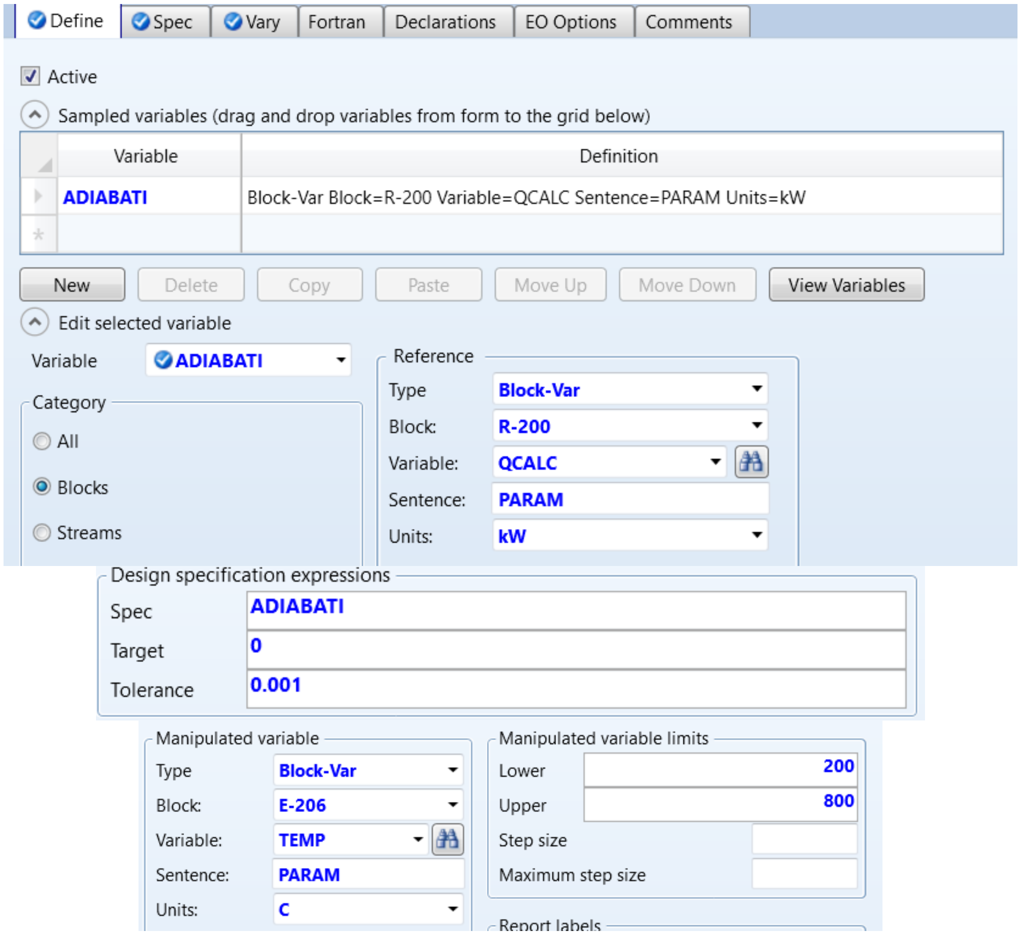


*Figure 12: Design Spec for incoming sand*

This calculator is used to change the temperature of the sand, exiting the heat exchanger.

| Variable | Information flow | Definition |
|---|---|---|
| TSAND1 | Import variable | Stream-Var Stream=HOTSAND2 Substream=CISOLID Variable=TEMP Units=C |
| TSAND2 | Export variable | Stream-Var Stream=HOTSAND1 Substream=NC Variable=TEMP Units=C |

Enter executable Fortran statements

```
      TSAND2=TSAND1
```

*Figure 13: Sand Calculator*

This calculator is used to calculate the temperature of the sand that is exiting E-206 and entering R-200. They are set equal since the HOTSAND2 is supposed to enter R-200 but again it makes our aspen crash, so it is separated.

**References:**

CHE 433: Process Simulation with Aspen Plus - Chapter 5 Physical Property Methods

CHE 433: Process Simulation with Aspen Plus - Chapter 6 Chemical Reactors and Non Conventional Solids

CHE 433: Process Simulation with Aspen Plus - Chapter 7 Distillation Unit Operations

ChatGPT